
BBM475 Management Information Systems
2025-26 Fall, Term Project on Business Intelligence
Due Dates: (T1) Nov 13, 2025, (T2: A & B) Dec 25, 2025

In this term project, you are expected to perform business intelligence (BI) on a corporate dataset that is provided by the lecturer (i.e. ISBSG dataset on software project estimation) or that you will find by yourself. Prior to using the ISBSG dataset, please read “Ch23-Project Planning” from the book *Software Engineering (9th Ed)* by Ian Sommerville.

If you will search for your own data source, then the data should be stored at an operational level (e.g., in a transaction processing system (TPS) or produced by a business process at that level) and it could be summarized and/or analyzed at higher management level(s), so that you are able to apply BI techniques at tactical level (i.e. middle level management) as expected in this project. You must first consult Lecturer to work on a different dataset.

You may work in groups of 2-3 people to carry out your project work. On the dataset that you have chosen, you are required to realize the following items per team member of your group:

- Raise **two** question to be answered by **MIS** and answer it by using BI techniques.
- Raise **one** questions to be answered by **DSS** and answer them by using BI techniques.

The project will consist of two stages and two deliveries (T1 & T2) at the end of these stages.

- To create T1, you must use Goal-Question-Metric approach [1] to design your BI project. You must provide designs of visual representations to answer the questions you raise in your T1 report.
- To create T2, you must apply BI techniques on the selected attributes of the dataset in order to create the visual representations that you have designed in T1. It is suggested that you use a BI tool such as Power BI [2] or Tableau [2], and use its features for data analytics and visualization.

The first delivery (T1) will be evaluated over 100, the second delivery (T2) will be evaluated over 200, and your final grade will be calculated by dividing the sum of the two by 3:

- (T1) will include your GQM tree only (including two questions for MIS and one question for DSS, per team member). T1 is not expected to include answers to the questions you raise; rather, it is a milestone to review, prior to putting further effort to answer the questions. **(Nov 13, 100p)**
- (T2) will include all details mentioned above, including the GQM tree and the answers to the questions in your tree. **(Dec 25, 200p)**

The grading of your project will be in parallel to the suitability of the questions for their specified purposes and the sophistication level of the BI techniques that you apply.

Enjoy your BI project!

References:

- [1] R. Van Solingen, V. Basili, G. Caldiera, D.H. Rombach, Goal Question Metric (GQM) Approach, in: Encyclopedia of Software Engineering, online vers.@Wiley Interscience, John Wiley & Sons, 2002.
- [2] <https://powerbi.microsoft.com/tr-tr/get-started/>
- [3] <https://www.tableau.com/trial/download-tableau/>